

Mid-Progress Report

Cardiovascular Disease Prediction with Explainable AI

Implementation and Adaptation of Shah et al. (2025)

1. Executive Summary

This mid-progress submission implements an explainable machine-learning pipeline for binary cardiovascular disease prediction. The work adapts the hybrid ensemble and explainability direction of Shah et al. (2025), but applies it to my selected cardiovascular disease dataset of 68,784 usable patient records with clinical and lifestyle features. The current implementation includes preprocessing, exploratory data analysis, feature engineering, six individual models, one voting ensemble, and four explainability techniques: SHAP, LIME, permutation importance, and partial dependence plots.

The strongest single AUC in the current run is CatBoost with ROC-AUC = 0.7994, while the voting ensemble gives ROC-AUC = 0.7987. The results are lower than the 0.82 AUC reported by Shah et al., but the output is still acceptable for the mid-progress stage because the full preprocessing, modeling, evaluation, and interpretability workflow is working end-to-end.

2. Difference Between Shah et al. and My Implementation

Aspect	Shah et al. (2025)	My Dataset / Notebook Implementation
Dataset	Combined public CVD datasets: IEEE Dataport Cardiovascular Disease Dataset, Cleveland Heart Disease Dataset, and Hungarian Dataset; about 70,000 instances and 12 features.	One cardiovascular disease dataset with 68,784 records and 12 original features; after feature engineering the modeling set contains 17 features.
Preprocessing	Missing values, IQR/outlier removal, Min-Max scaling, 80/20 stratified split, and imbalance treatment using SMOTE plus random undersampling.	Missing-value checking, statistical summary, outlier-aware analysis, StandardScaler normalization, 80/20 stratified split, and SMOTE balancing.
Feature engineering	Engineered features such as BMI, cholesterol-glucose ratio, and blood-pressure interactions.	Created BMI, BP_RATIO, PULSE_PRESSURE, AGE_BMI_INTERACTION, and BP_CHOL_INTERACTION.
Model design	Stacked ensemble: Gradient Boosting, CatBoost, LightGBM, SVM, and Neural Network as base models, with XGBoost as meta-learner.	Trained XGBoost, LightGBM, CatBoost, Random Forest, Gradient Boosting, and Logistic Regression; combined them using soft Voting Ensemble.
Explainability	Mainly SHAP, with PCA/t-SNE visualizations for multidimensional interpretation.	SHAP, LIME, permutation importance, and PDP were implemented. This is broader for XAI tools, but PCA/t-SNE were not yet included.
Performance	Reported AUC-ROC about 0.82, Precision 81%, Recall 83%, and F1 82%.	Best AUC is about 0.7994. Current results are close but lower; improvement can come from hyperparameter tuning, better feature selection, and possibly a true stacking ensemble.

3. Work Completed So Far

- Data preprocessing:** Loaded the dataset, checked missing values, reviewed descriptive statistics, prepared features and target, applied scaling, train-test split, and SMOTE balancing.
- EDA:** Generated class distribution, age distribution, blood pressure scatter plot, cholesterol-glucose plot, and a full correlation heatmap.
- Feature engineering:** Added clinically meaningful derived variables: BMI, BP ratio, pulse pressure, age-BMI interaction, and blood-pressure/cholesterol interaction.
- Model building:** Implemented six individual models and one ensemble model. Each model was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, ROC curves, and confusion matrices.

- **Explainability:** Implemented SHAP summary bar, SHAP beeswarm, SHAP force plots, LIME explanations for positive and negative cases, permutation importance, and PDP plots.

4. Preliminary Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.7290	0.7455	0.6870	0.7150	0.7975
LightGBM	0.7295	0.7458	0.6879	0.7157	0.7984
CatBoost	0.7299	0.7473	0.6863	0.7155	0.7994
Random Forest	0.7283	0.7458	0.6842	0.7137	0.7977
Gradient Boosting	0.7278	0.7427	0.6885	0.7145	0.7970
Logistic Regression	0.7234	0.7385	0.6829	0.7096	0.7876
Voting Ensemble	0.7290	0.7458	0.6864	0.7149	0.7987

Interpretation: The tree-boosting models have very similar performance. CatBoost gives the highest ROC-AUC, while LightGBM has the highest accuracy among the individual models. The voting ensemble is stable but does not clearly outperform CatBoost, which means that Phase 3 should test a true stacking ensemble with XGBoost as meta-learner to better match the paper.

5. Explainability and Clinical Interpretation

The global explanation results consistently show that blood pressure is the dominant risk signal. SHAP and permutation importance both rank AP_HIGH as the most influential feature. The engineered BP_CHOL_INTERACTION is also important, meaning the model benefits from combining systolic blood pressure with cholesterol category. AGE and AGE_BMI_INTERACTION contribute additional risk information. PDP plots show that higher AP_HIGH generally increases predicted disease probability, while interaction features help capture non-linear risk patterns.

LIME was used to explain individual positive and negative predictions. This is useful because it shows which features pushed one patient-level prediction toward cardiovascular disease or away from it. However, LIME results should be presented as local explanations only; they should not replace global feature importance.

6. Challenges, Limitations, and Next Steps

Current challenge	Planned action for final phase
The current ensemble uses soft voting, while the paper uses a stacking design with XGBoost as meta-learner.	Implement StackingClassifier or manually create meta-features and train XGBoost as the final meta-model.
Current AUC is around 0.80, lower than the paper's reported 0.82.	Tune hyperparameters using GridSearchCV or Optuna and test feature selection.
PCA/t-SNE from the paper are not yet included.	Add PCA/t-SNE visualizations or clearly state that this implementation uses LIME and PDP instead.
README still contains some expected/example metrics that do not match the actual notebook output.	Replace expected results with the actual table reported above and update the file organization to match the submitted repository.

7. What Must Be Included in the Submission

- A separate EDA/preprocessing notebook shared by the team, plus this individual model notebook named with the student name.
- The GitHub repository link, with organized folders for notebooks, results, explainability outputs, and README.
- The results table, confusion matrices, ROC curves, SHAP plots, LIME HTML files, permutation importance, and PDP plots.
- A short discussion of how the paper was adapted, not copied exactly.
- A limitations/challenges section and clear next steps for Phase 3.

References

Shah, P., Shukla, M., Dholakia, N. H., & Gupta, H. (2025). Predicting cardiovascular risk with hybrid ensemble learning and explainable AI. *Scientific Reports*, 15, 17927. <https://doi.org/10.1038/s41598-025-01650-7>

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